

MODULE 8 — Integrations

SRS v1.0 | Status: Draft → Pending Your Confirmation

1. Purpose

This module defines **how the platform connects to the outside world**—cleanly, securely, and without leaking control.

The guiding principle here is **centralized power, controlled exposure**:

- Platform admin owns core infrastructure integrations
- Customers consume capabilities, not credentials

This protects margin, reliability, and compliance.

2. Scope

Applies to:

- Telephony integrations (platform-managed)
- Calendar synchronization
- Webhooks (outbound & inbound)
- Internal vs external integration boundaries

Does **not** include:

- Customer-managed telephony providers
 - Third-party CRM sync
 - Automation marketplaces (Zapier / n8n)
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3. Telephony Integration (Platform-Managed)

3.1 Ownership Model (Confirmed)

- Telephony providers are:
 - Integrated
 - Configured
 - Maintained**only by Platform Admin**
- Customers **do not** supply provider credentials.
- Customers **select or purchase numbers** from available inventory.

This ensures:

- Cost control
 - Uniform call quality
 - Central compliance enforcement
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3.2 Phone Number Management

Platform Admin capabilities:

- Procure numbers from providers
- Assign numbers to tenants
- Configure:
 - Inbound routing
 - Outbound caller ID
 - Regional availability
- Set pricing per number (optional)

Tenant capabilities:

- View available numbers
 - Purchase / assign number(s)
 - Release numbers
 - View cost implications per number
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3.3 Cost Attribution

- Each number is tagged with:
 - Region
 - Cost per minute (inbound / outbound)
- Call cost calculation considers:
 - Number used
 - Direction

- Duration
- Costs feed directly into:
 - Usage metering
 - Analytics
 - Billing (Module 9)

No flat-rate masking. Full transparency.

3.4 Routing Rules

Inbound routing supports:

- AI-first handling
- Time-based routing
- Language-based routing
- Human fallback

Outbound routing supports:

- Campaign-driven number selection
 - Region matching
 - Compliance constraints
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4. Calendar Integration (Confirmed)

4.1 Supported Calendars

- Google Calendar
 - Microsoft Outlook Calendar
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4.2 Use Cases

- AI scheduling appointments
 - Availability checking during calls
 - Booking confirmations
 - Rescheduling / cancellation
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4.3 Sync Behavior

- Two-way sync
- Conflict detection
- Timezone normalization
- Event ownership mapping

Calendar sync must be:

- Explicitly authorized
 - Revocable
 - Logged
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4.4 AI Boundaries

AI may:

- Check availability
- Create events
- Update events

AI may **not**:

- Delete calendar accounts
 - Access unrelated calendars
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5. Webhooks

5.1 Outbound Webhooks (Confirmed)

Platform emits webhooks for:

- Lead created / updated
- Call completed
- Campaign completed
- Deal stage changed
- Appointment booked

Webhook payloads are:

- Versioned

- Signed
 - Retry-enabled
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5.2 Inbound Webhooks (Controlled)

Inbound webhooks allowed for:

- Lead creation
- Lead updates
- Campaign triggers

Constraints:

- Authentication required
- Rate-limited
- Schema-validated

No arbitrary execution allowed.

6. Internal vs External Integration Boundaries

6.1 Internal Integrations

Internal-only:

- Telephony providers
- AI engines
- Billing meters
- Usage trackers

These are **never exposed** to customers.

6.2 External Integrations

Customer-visible:

- Calendar connectors
- Webhooks
- Export APIs (read-only, analytics)

Customers interact with:

- Abstracted endpoints
 - Controlled schemas
 - Limited scopes
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7. API Exposure Rules

- No raw provider credentials exposed
- No write access to billing or telephony
- All APIs are:
 - Tenant-scoped
 - Permission-aware
 - Rate-limited

API keys:

- Generated by platform
 - Rotatable
 - Revocable
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8. Error Handling & Resilience

- Provider failures must:
 - Fail gracefully
 - Trigger alerts
 - Pause dependent workflows
- Automatic retries with backoff
- Circuit breakers for unstable providers

No cascading failures allowed.

9. Audit & Observability

- Log:
 - Integration calls
 - Failures
 - Authorization events

- Tenant visibility:
 - Success/failure counts
 - Admin visibility:
 - Provider health
 - Latency
 - Error rates
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10. Non-Functional Requirements (Module-Specific)

- Webhook delivery SLA $\geq 99.9\%$
 - Calendar sync latency ≤ 10 seconds
 - Telephony routing decision $\leq 200\text{ms}$
 - Integration outages must not corrupt CRM data
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11. Out of Scope (Explicit)

- Customer-provided telephony credentials
 - Third-party CRM integrations
 - Marketplace connectors
 - Custom customer scripts
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